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From: Rehan Edirisinghe

Subject: Evaluation of vaccine storage unit

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FOREWORD

Soluvax, Inc. is a global health technology company focused on developing solar-powered vaccine refrigeration systems for rural and off-grid clinics. Soluvax's mission is to ensure safe and reliable cold chain storage for vaccines in remote areas with unstable electricity supplies. The company has approached R Engineering Services seeking technical assistance to test and characterize the vapor-compression cycle under different compressor speeds. The company has provided a prototype unit using refrigerant R-134a mounted to a Hampden H-CRT-1 refrigeration trainer that is already instrumented to measure pressure, temperature, and flow rate. Soluvax requested that R Engineering Services use the fully instrumented rig to document temperature-specific entropy (T-s) diagrams for the unit with heat exchange to and from the ambient environment while the evaporator and condenser fans operate at full airflow (100% speed) and the compressor operates at angular speeds of 45 Hz, 52 Hz, and 60 Hz. Additionally, Soluvax, Inc. requested that R Engineering Services report the system's cooling capacity, coefficient of performance, and energy efficiency ratio for the system at each speed setting. Lastly, Soluvax requests an analysis on how the compressor speed can be controlled to reduce power demand while maintaining vaccine safety concurrently. This report outlines the testing methods, results, and analysis of the experimental data, along with the recommendations for the subject CFTAH, prepared by R Engineering Services.

SUMMARY

R Engineering Services conducted physical testing on the provided Refrigeration system mounted to a Hampden H-CRT-1 refrigeration trainer. The system's operating metrics were quantified through T-s diagrams created using the physical testing data at a steady state. We identified a correlation between the cooling capacity of the system has an apparent correlation with the compressor speed. The correlation is expected based on theory and shows that lowering the compressor speed to 45Hz (cooling capacity of 760.1 $\pm$ 15 W at an energy efficiency ratio of 1.98 $\pm$ 0.05) is favorable compared to running the compressor at 52Hz (cooling capacity of 771.7 $\pm$ 15 W at an energy efficiency ratio of 1.92 $\pm$ 0.05) when energy efficiency is imperative. On the flip side, if cooling is to be prioritized, a higher compressor speed of 59 Hz results in a cooling capacity of 802.1 $\pm$ 15W is required, albeit at the cost of energy efficiency with a reported energy efficiency ratio of 1.82 $\pm$ 0.05.

Based on our analysis and understanding of the functional requirements of the cold chain vaccine storage, we recommend a control that involves short-term weather forecasts along with a real-time compressor speed adjustment based on current solar irradiance. Concurrently, the proposed system would exploit the full acceptable temperature range depending on energy availability while adhering to CDC requirements for vaccine storage.

1.0 METHODS

As requested by Soluvax, R Engineering Services performed testing on the provided fully instrumented prototype unit using refrigerant R-134a mounted to a Hampden H-CRT-1 refrigeration trainer to gather pressure, temperature, power usage, and flow rate data at different compressor speeds. This section outlines the procedure we used to characterize the vapor-compression cycles by processing the data obtained from physical testing. Additionally, this section outlines the method used to determine the cooling capacity, coefficient of performance, and energy efficiency ratio of the system at each compression speed. Together, these experiments can be used to identify the ideal operating conditions for the refrigeration system in energy-poor environments.

1.1 Physical Testing

Physical testing on the subject refrigeration equipment attached to the Hampden H-CRT-1 refrigeration trainer was completed in a controlled environment with a known atmospheric pressure and ambient temperature. This value was taken from refrigerant R-134a, which was used in the experiment. **Figure 1** is an annotated image and simplified diagram of the setup used to identify the T-s diagrams of the subject system at different compressor speeds. Note that, although the experiment is conducted in the ambient environment, we will refer to the "cold" and "hot" reservoirs as the immediate environments surrounding the evaporator and condenser, respectively. In a refrigerator, the cold reservoir would be the interior of the refrigerator, and the hot reservoir would be the outside.

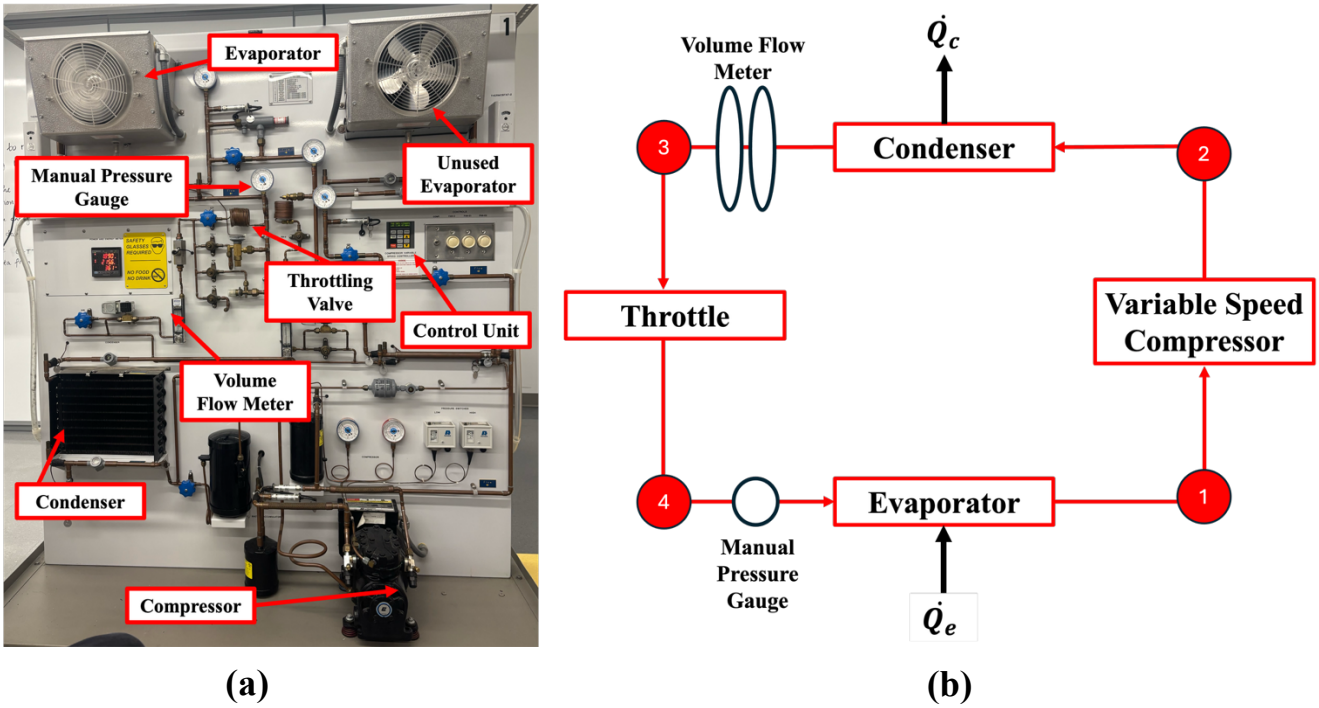


Figure 1. Annotated Experimental setup with an associated simplified diagram. The experimental setup utilized the provided prototype unit, mounted on a Hampden H-CRT-1 refrigeration trainer. **(a)** The labelled experiment setup on the Hampden H-CRT-1 refrigeration trainer. **(b)** Simplified diagram of the refrigeration cycle used in the experimental setup. Numbers 1-4 represent different states of R-134a. The compressor was controlled using the control unit, and the evaporator and condenser were set to 100% airflow. A digital pressure and temperature sensor was attached to the inlet and outlet of the compressor, condenser, and evaporator, with the only exception being a physical pressure sensor connected to the inlet of the evaporator instead of a digital one. The volume flow meter was used as a qualitative measure for the state of R-134a between the throttle and condenser. $\dot{Q}_e$ represents the rate of heat absorbed from the cold reservoir, and $\dot{Q}_c$ represents the rate of heat expelled into the warm reservoir.

The control unit was used to adjust the evaporator, condenser, and compressor speeds. The evaporator and condenser fans were set to a constant 100% airflow throughout the experiment, while the compressor speed was varied between 45Hz and 59Hz. Although Soluvax's task letter specifies a maximum compressor speed of 60Hz, this value exceeded the maximum threshold of the experimental setup, which was 59Hz. The data was recorded using LabVIEW, a programmable data acquisition system by National Instruments. A custom program was designed to collect data on temperature, pressure, flow rate, and electrical power sensors from the refrigeration system.

Once the compressor speed was set, the subject refrigeration unit was allowed to run until it reached a steady state. We define steady state as the point at which the pressure and temperature at each point in the cycle become constant, which, in practical terms, means that the cold reservoir (the refrigerator interior) is no longer cooling further. Once this state was reached (on average, about 30 minutes), the data from the pressure, temperature, flow rate, and power sensors were recorded.

1.2 Obtaining the T-s diagrams

To characterize the temperature-specific entropy (T-s) diagrams for the refrigeration unit, we processed the steady-state data obtained from the physical experiments in MATLAB. Within MATLAB, we used Coolprop, an open-source library for solving thermodynamic cycles of a given fluid, in this case, refrigerant R-134a. Vapor dome data of R-134a was used, and Coolprop was used to classify the associated entropy for an associated pressure and temperature at each stage in the cycle. It is important to note that only six states are on the T-s diagram. We operated under the assumption that there are no losses between the condenser outlet and throttle inlet, as well as the throttle outlet and evaporator inlet. The points were plotted against the vapor dome associated with refrigerant R-134a.

1.3 Cooling Capacity, Coefficient of Performance & Energy Efficiency Ratio

Cooling capacity is defined as the rate at which heat is absorbed from the cold reservoir. To find the cooling capacity, the mass flow rate needed to be determined first. Using the volumetric flow rate data of the system and the density of the refrigerant R-134a at a specific state using Coolprop, we can calculate the mass flow rate using **Equation 1**. The volume flow meter must be attached to the condenser outlet because R-134a exits the outlet as a saturated liquid, ensuring an accurate flow rate is recorded.

$$\dot{m} = \dot{V} * \rho \qquad \textbf{(1)}$$

Using **Equation 1** and the enthalpies calculated using Coolprop and MATLAB, the cooling capacity can be calculated using **Equation 2** [1]. The cooling capacity is defined as the rate at which heat is absorbed by the evaporator from the cold reservoir, or the difference in enthalpies between the inlet and outlet of the evaporator multiplied by the mass flow rate.

$$\dot{Q}_e = \dot{m}(h_1 - h_4) \qquad \textbf{(2)}$$

After finding the cooling capacity of the system, it can be used to find the coefficient of performance (hereafter, "COP") of the system. The COP is the ratio of the cooling capacity to the work done by the compressor into the

system. We used **Equation 3** to calculate the effective rate of work by the compressor into the system where $h_2 - h_1$ refers to the difference in enthalpies between the compressor inlet and outlet.

$$\dot{W_{COMP}} = \dot{m}(h_2 - h_1) \quad \textbf{(3)}$$

Once the cooling capacity and effective rate of work by the compressor are calculated, the COP is computed using **Equation 4** [1].

$$\beta = \text{COP} = \frac{\dot{Q}_e}{\dot{W}_{\text{comp}}} \quad \textbf{(4)}$$

Because the compressor is not 100% efficient, a useful performance measure is the energy efficiency ratio (hereafter, "EER"). The EER, **Equation 5**, compares the cooling capacity to the electrical power supplied to the system, rather than the rate of compressor work used in the COP calculation.

$$EER = \frac{\dot{Q}_e}{\dot{W}_{\text{elec}}} \quad \textbf{(5)}$$

Altogether, each of these metrics was calculated at each of the compressor speeds to understand the operational benefits and downfalls of increasing or decreasing the compressor speed from the current rate of 52Hz in use by Soluvax.

1.4 Error Analysis

Because the entropy and enthalpy are values extracted from the Coolprop database, calculating the error for the T-s diagram and the cooling capacity deviated from conventional methods. Instead, the perturbation method was used to calculate errors in the entropy and enthalpy values. This was done by perturbing the input temperature and pressure by their respective errors and observing the sensitivity of the enthalpy and entropy values. The derived enthalpy error values were used in finding the error in the compressor's work and the system's cooling capacity, and therefore the error in the COP and EER.

2.0 RESULTS

The objective of these experiments was to identify the T-s diagrams of the refrigeration cycle at different compressor speeds and understand the performance implications of varying the compressor speed. The results will help shape the operation of Soluvax's cold-chain storage for vaccines in remote areas where the electricity supply is unstable.

2.1 T-s Diagrams

The experiments were conducted in a controlled laboratory environment with a known atmospheric pressure of 98.2 ± 0.05 kPa and an ambient temperature of 22.4 ± 0.1 °C. The experiments were conducted as described in **Section 1.1** and yielded temperature and pressure data at 6 points in the refrigeration cycle once steady state was reached. Using MATLAB and CoolProp, we determined the associated temperature-specific entropy at

each inlet and outlet and plotted the values against the vapor dome of refrigerant R134a. The result is a 6-point plot shown in **Figure 2**.

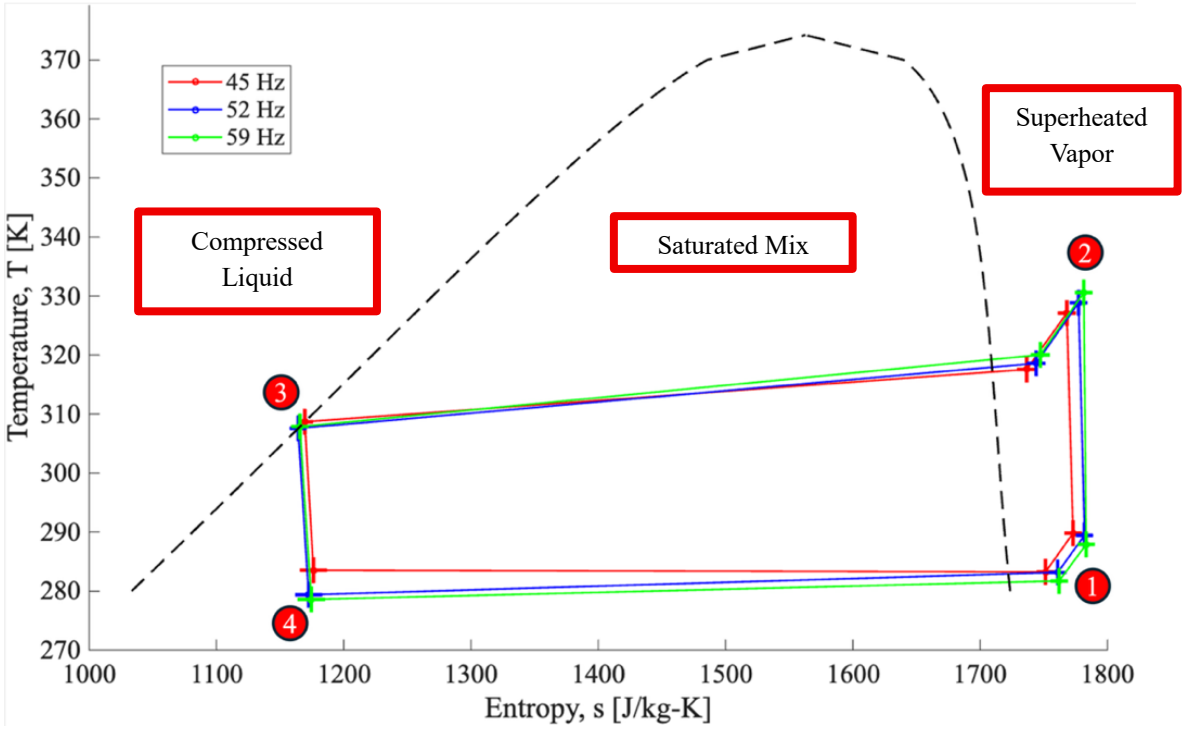


Figure 2. Combined Temperature-specific entropy (T-s) diagram of the three compressor speeds tested with error bars. The vapor dome was found using Coolprop's database for R-134a. Each of the colored lines represents a different compressor speed, and each point represents a different point in the cycle. We numbered each point in correspondence with **Figure 1 (b)** to clarify the state of the refrigerant at each stage in the cycle. The dominant source of error is the accuracy error in the temperature and pressure values, which is propagated into the entropy value.

The decision not to assume that there are no losses between components in the refrigeration cycle proved essential based on the results. By measuring both the inlet and outlet of every component (except the throttle), an apparent mismatch was found in the T-s diagrams, as illustrated by two distinct points at states 1 and 2 in **Figure 2.** If the assumption were made, there would be hidden discrepancies in the model.

2.2 Performance Metrics

After plotting the T-s diagrams, the cooling capacity was found using the mass flow rate, derived from the volume flow rate data collected during the experiments. Using the cooling capacity, we determined the COP and EER at each compressor speed by applying **Equation 4** and **Equation 5**. **Table 1** summarizes the cooling capacity, COP, and EER of the system at each compressor speed.

Table 1. Cooling Capacity, Power consumption, COP, and EER at each compressor speed. The dominant sources of error originate from the fact that each performance metric relies on derived quantities from the original temperature, pressure, power, and flow rate data. Specifically, the error in the COP is an order of magnitude higher than the EER because the COP relies on two derived quantities, cooling capacity and work done by the compressor (derived quantities that rely on computed enthalpy values from different states). In contrast, EER has one derived quantity (cooling capacity) and one raw measurement (electrical power supplied).

Compressor Speed	Cooling Capacity [W]	Power Consumption [W]	COP	EER
45Hz	760.1 $\pm$ 15	388.34 $\pm$ **2**	6.785 $\pm$ **0.8**	1.98 $\pm$ 0.05
52Hz	771.7 $\pm$ 15	402.85 $\pm$ **2**	6.335 $\pm$ 0.7	1.92 $\pm$ 0.05
59Hz	802.1 $\pm$ 15	440.77 $\pm$ **2**	5.701 $\pm$ 0.6	1.82 $\pm$ 0.05

Table 1 shows a correlation between cooling capacity and compressor speed. As the compressor speed increases, the cooling capacity also increases, as expected by the theory. Additionally, another trend of interest is that the COP also appears to have a direct correlation with the compressor speed. However, once the electrical power input is considered in the EER, a higher compressor speed of 59Hz is significantly less efficient than a lower one of 45Hz.

3.0 DISCUSSION

Understanding the capabilities and performance of the subject refrigeration equipment is crucial for identifying the optimal operation strategy for safely refrigerating vaccines while relying primarily on solar energy. Ensuring that the system operates efficiently, utilizing the limited energy while maintaining a refrigeration temperature of 2-8 °C [2], is imperative to delivering Soluvax's mission of increasing vaccine availability.

3.1 Operational Requirements at Different Compressor Speeds

As expected by the theory of the refrigeration cycle [1], the performance metrics of the subject refrigeration unit exhibit a correlation between the system's cooling capacity and the compressor speed. However, the increased cooling capacity comes at the cost of increased electrical power usage and a reduction in the EER, which is the most important metric when considering the operating environment of the vaccine cold chain storage.

3.2 Optimized Operating Strategy

Given the apparent correlation between cooling capacity and compressor speed in our limited data set, the most obvious recommendation is that running the compressor at a speed of 59 Hz is beneficial when there is significant access to electrical power. However, the opposite is true when electrical power is limited because there is an energy and efficiency penalty to running the compressor at high speeds. The results suggest that despite the increase in electrical power, the heat energy extracted from the refrigeration does not increase by the same amount. R Engineering Services recommends further testing at a broader range of compressor speeds, which could be used to find the compressor speed associated with the peak EER to identify the most efficient compressor speed to run the cold chain storage at during periods of limited energy access.

Because CDC guidelines on vaccine storage requirements of 2-8 °C [2], an optimal strategy would be to exploit the full acceptable range. For example, cooling the refrigerator interior to 2 °C during periods of high solar irradiation and maintaining a temperature of 8 °C during periods of limited solar irradiance would be an effective strategy that maximizes cooling during periods of high energy availability while conserving it when supply is low. Combined with adequate insulation, the refrigerator can be allowed to warm up to just below the upper limit of vaccine safety, 8 °C, slowly and maintained at that temperature during periods of low solar irradiance.

Further optimization may be possible through weather forecast-based control in the cooling strategy for the cold-chain storage. For example, methods used in PV maximum power point tracking (MPPT) [3] demonstrate how power systems can optimize themselves to varying solar irradiances. Similar principles can be applied to control the speed of the cold chain storage's compressor. Altogether, a vaccine cold chain storage solution with an attached PV system could be heavily optimized by controlling the compressor speed based on the available solar energy, while also considering weather forecasts to prepare for different operating environments.

4.0 CONCLUSIONS & RECOMMENDATIONS

Based on the physical tests and analyses conducted on the subject refrigeration unit attached to the Hampden H-CRT-1 refrigeration trainer provided by Soluvax, we can conclude that, within the bounds of experimental uncertainty, the cooling capacity of the system has an apparent correlation with the compressor speed. These results match what is expected based on theory and show that lowering the compressor speed to 45Hz is beneficial (cooling capacity of 760.1 $\pm$ 15 W at an EER of 1.98 $\pm$ 0.05) compared to the previously mentioned operating speed of 52Hz (cooling capacity of 771.7 $\pm$ 15 W at an EER of 1.92 $\pm$ 0.05) when energy efficiency is imperative. On the other hand, if cooling is to be prioritized, a higher compressor speed of 59 Hz results in a cooling capacity of 802.1 $\pm$ 15W is required, albeit at the cost of energy efficiency with an EER of 1.82 $\pm$ 0.05.

Given the functional requirements of the subject cold-chain system, an optimized energy-use and temperature-control strategy to maintain vaccines within the required window is necessary. R Engineering Services recommends a controls approach that combines short-term weather forecasting with compressor speed control based on real-time solar intensity. Under such a system, the compressor runs at its highest setting to cool the vaccines to the minimum acceptable temperature of 2 °C during daytime or times of high solar irradiance. By the same token, during times of low solar irradiance (i.e., nighttime), the system would allow the storage to rise to the maximum allowable temperature of 8°C and then maintain it. This strategy delays the compressor startup and energy usage as a result.

Lastly, R Engineering Services recommends further testing on the cold-chain storage system to find the optimal compressor speed that is associated with the highest EER, and also explore the potential implications of varying the condenser and evaporator fan speeds to unlock further energy savings.

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